SMART LOST AND FOUND MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

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RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "SMART LOST AND FOUND MANAGEMENT SYSTEM" is the bonafide work of DARSHAN ABINAV RK (241501039), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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DECLARATION

I, DARSHAN ABINAV RK (241501039), hereby declare that the project report titled "SMART LOST AND FOUND MANAGEMENT SYSTEM", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report, including its data model, forms, reports, page, Deluge scripts and workflows, was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature, code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: _____

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DARSHAN ABINAV RK

ABSTRACT

Small personal belongings such as books, calculators, identity cards, wallets and keys are lost and found on a college campus every day, yet there is rarely a proper record of them. Reports travel on messaging groups or by word of mouth, finders hand items to whoever is nearby, and ownership is checked by a few spoken questions. Owners cannot tell whether their item has been found, finders cannot show that it was reported, and the administrator has no count of what is waiting to be claimed.

A low-code approach on Zoho Creator suits this problem because it is a data-capture and notification problem that forms, reports, workflows and Deluge scripting solve directly, with hosting, security and mobile access supplied by the platform. The Smart Lost and Found Management System has three forms, Lost_Items, Found_Items and Claim_Requests, linked by a lookup, with a subform that records the actions taken on each claim. Five list reports and a dashboard page present the data. Seven Deluge form workflows reject future dates, stop duplicate lost reports, email the administrator when an item is found or a claim is submitted, and set a found item to Claimed when its claim is approved. Admin and Student permission profiles control access, and the application also opens in the Zoho Creator mobile app.

The application was tested in the Development environment with sample data of 10 lost items, 11 found items and 12 claim requests. The future-date message on the claim form, the duplicate-report message and the claim notification email were observed working as designed. Testing also exposed four open defects, chiefly two approved claims whose found items are not marked Claimed and dashboard totals that do not match the reports. Deployment to Production, a login test for the Student profile and a user acceptance test remain for future work.

Keywords: Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Lost and Found Management; Permission Profiles.

PROJECT AT A GLANCE

Attribute	Details
Project Title	Smart Lost and Found Management System
Application Domain	Education – campus lost and found management
Student Name & Register No.	DARSHAN ABINAV RK – 241501039 (Reg. No. 2116241501039)
Course	CB23F35 – Low-Code Application Development using Zoho Creator
Zoho Creator Edition	Trial edition
Application Name	Smart Lost and Found Management System
Published / Live URL	Not published to Production; the application runs in the Development environment of Zoho Creator
Demo Credentials (per role)	Administrator: application owner account, shared at the demonstration. Student: profile configured; login test not yet carried out
Number of Forms / Subforms	3 forms / 1 subform
Number of Reports (by type)	5 reports, all List type: Lost Items Report, All Items, Found_Items Report, Claim Requests Report, All Claim Requests
Number of Pages / Dashboards	1 page: Lost and Found Dashboard
Number of Workflows	7 form workflows (4 on validation, 3 on successful submission); schedule 0, approval 0, Blueprint 0, batch 0
Deluge Functions / Custom Actions	No custom function; 7 Deluge workflow scripts
Integrations	Zoho Creator email notifications; no external REST API
Profiles Defined	Permission profiles: Admin, Student
Environments Used	Development
Date of Production Deployment	Not deployed

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
AIML	Artificial Intelligence and Machine Learning
API	Application Programming Interface
CO	Course Outcome
CRM	Customer Relationship Management
DS	Deluge Script (identifier)
ER	Entity–Relationship
FR	Functional Requirement
ID	Identifier (record ID)
LCAP	Low-Code Application Platform
NFR	Non-Functional Requirement
OCR	Optical Character Recognition
REST	Representational State Transfer
TC	Test Case
UAT	User Acceptance Testing
UC	Use Case
URL	Uniform Resource Locator
Zia	Zoho’s AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Every day thousands of students and staff move between classrooms, laboratories, the library, the canteen, bus stops and playgrounds on a college campus. Along the way, small belongings such as books, calculators, USB drives, identity cards, wallets, keys and water bottles are left behind. Two people are then involved: the owner, who wants the item back, and the finder, who wants to return it but often does not know whom to approach.

In most colleges this is handled informally. The owner asks classmates, a message is forwarded on a messaging group, or the finder leaves the item with whichever staff member is nearby. No single place holds the lost reports, the found reports and the ownership claims together. The Smart Lost and Found Management System described in this report is a business application built on the Zoho Creator low-code platform for a campus such as Rajalakshmi Engineering College. It records lost items, found items and claim requests, checks the data entered, informs the administrator by email, marks a found item as Claimed when its claim is approved, and presents the position on a dashboard.

1.2 Problem Statement

Lost and found handling on a campus depends on memory, conversation and scattered messages rather than on a maintained record. An owner cannot easily learn whether a matching item has been handed in, a finder cannot easily show that an item was reported, and the administrator cannot easily see how many items are waiting to be claimed. The problem addressed here is to provide one controlled, searchable and partly automated record of lost items, found items and the claims made against found items.

- **No central record:** lost and found details sit in separate chats, registers or in people’s memory, so the same item may be reported several times and never matched.
- **No proof or history:** ownership is checked by spoken questions, and nothing records who claimed an item, on which date and with what proof.
- **Delayed awareness:** the administrator learns of a found item or a claim only when somebody happens to mention it.
- **Invalid data:** dates in the future, missing contact details and repeated reports enter the record because nothing checks them.
- **No visibility and no access control:** nobody can see how many items are lost, found or pending, and anyone who holds the register can alter it.

1.3 Objectives

Each objective below is traced to a module in Chapter 5 and to a result in Chapter 8.

- To design a data model and forms for lost items, found items and claim requests, with a lookup from claims to found items and a subform for the actions taken on a claim (Sections 4.2, 4.3 and 5.2).
- To automate date validation, duplicate detection, administrator notification and status updating using Deluge scripts and form workflows (Sections 5.5 and 5.6).
- To provide a report for every form and a dashboard that summarises claims and items (Sections 5.3 and 5.4).
- To control access through Admin and Student permission profiles (Section 5.8).
- To make the application usable from a web browser and from the Zoho Creator mobile app (Section 5.11).
- To test the application and to record its deployment status, security controls, defects and limitations honestly (Chapters 6, 7 and 8).

1.4 Scope of the Project

In scope. The application has three forms (Lost_Items, Found_Items and Claim_Requests) and one subform inside Claim_Requests. It provides five list reports, one dashboard page, seven form workflows written in Deluge and two permission profiles (Admin and Student). Notifications reach the administrator through the Zoho Creator email action, and the application can be used on the web and in the Zoho Creator mobile app.

Out of scope. The application does not match lost items to found items automatically, does not connect to Zoho CRM, a payment service or any external REST API, has no customer portal and has not been published to a Production environment. Scheduled reminders, approval processes and Blueprint stages were not built. Section 9.2 lists these as future enhancements.

1.5 Justification for a Low-Code Approach

A low-code application platform lets a developer build data-driven business applications mainly through visual configuration and to add code only where the logic needs it [3]. Zoho Creator suits this problem for four reasons. First, the task is one of data capture and workflow, and forms, reports, validations and notifications are exactly the building blocks the platform provides [1]. Second, hosting, security, backups and mobile access come with the platform, so no server and no separate mobile application had to be developed. Third, the logic required, such as comparing a date with today’s date or sending an email after a record is saved, takes only a few lines of Deluge that a citizen developer can maintain. Fourth, low-code application platforms are treated by industry analysts as a distinct and maturing product category [2], so the skills practised here are useful beyond this project.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and the related literature. Chapter 3 gives the requirement analysis and Chapter 4 the system design. Chapter 5 describes the implementation

on Zoho Creator and Chapter 6 covers testing and debugging. Chapter 7 deals with deployment, performance, security and governance. Chapter 8 presents the results and discussion, and Chapter 9 concludes with future enhancements. The appendices hold the feature checklist, the course outcome mapping, the complete Deluge code and the user manual, followed by the examiner’s rubric and viva record sheet.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

On most campuses the lost and found process is manual. A student who loses an item asks friends, retraces the day's route, or posts a message on a notice board or a messaging group. A finder either keeps the item or hands it to a staff member such as a security guard, a department assistant or a faculty member. Whether the finder's name and the description of the item are written down depends entirely on the person who receives it.

When the owner finally reaches the person holding the item, ownership is usually settled by a few spoken questions. The item is handed over and the matter ends with no record of who took it. Figure 2.1 shows this as-is flow.

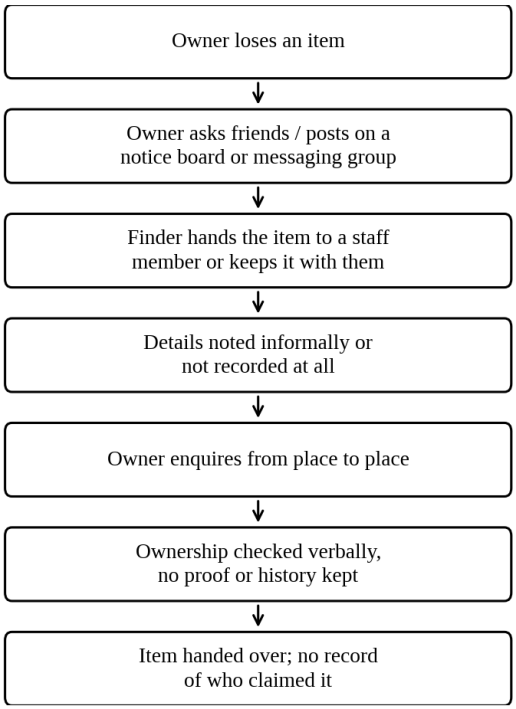


Figure 2.1: As-Is Process Flow of the Existing Lost and Found Practice

2.2 Limitations of the Existing System

- Lost reports and found reports are never held together, so a match happens by chance.
- Item details, dates and locations are not written in any consistent format.
- A claim leaves no evidence, so a wrong handover cannot be traced afterwards.
- The staff member in charge is not told automatically when an item is found or claimed.
- There are no counts, reports or dashboards, so unclaimed items cannot be reviewed.
- Where records exist, anybody can alter them because no access control is applied.

2.3 Low-Code versus Traditional Development

Table 2.1 compares the two approaches for this particular application. The effort figures are indicative estimates for a single developer, made to show relative scale only; no measurement was taken.

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Estimated 6 to 8 weeks for forms, database, back end, notification service and dashboard	Estimated 2 to 3 weeks; forms, reports and dashboard are configured visually
Skills required	Web front end, server language, database design and email integration	Data modelling, form design and basic Deluge scripting
Hosting and infrastructure	Server, database and email service must be arranged and maintained	Provided by the Zoho cloud
Mobile application	Separate mobile development or a responsive redesign	The Zoho Creator mobile app opens the same application (Figure 5.13)
Security and access control	Login, roles and permissions must be coded	Permission profiles configured in the Users section
Maintenance and change	Code change, testing and redeployment for every change	A field, report or validation is edited in the builder and tested at once
Customisation limits	Practically unlimited	Limited to platform features; pro-code extensions are needed beyond them
Cost	Developer time plus hosting cost	Platform subscription; a trial edition was used for this project

2.4 Literature Survey

Eight sources were reviewed: the vendor documentation, an industry analyst report, five research papers on low-code platforms and the Deluge language reference. Table 2.2 summarises them and states their relevance to this project.

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source and Year	Title / Product	Key Findings	Relevance to this Project
1	Zoho Corporation [1]	Zoho Creator Help Documentation	Describes forms, reports, pages, workflows, users and deployment in Zoho Creator.	Primary reference for configuring the application.
2	Gartner, Inc. [2]	Magic Quadrant for Enterprise Low-Code Application Platforms	Evaluates vendors in the low-code application platform market.	Shows that low-code platforms are an established category.
3	Bock and Frank [3], 2021	Low-Code Platform	Defines the low-code platform concept and what separates it from earlier approaches.	Supports the definition of low-code used in Chapter 1.
4	Sahay et al. [4], 2020	Supporting the Understanding and Comparison of Low-Code Development Platforms	Proposes a feature-based way to compare low-code platforms.	Guides the feature checklist in Appendix A.

S.No.	Author(s) / Source and Year	Title / Product	Key Findings	Relevance to this Project
5	Luo et al. [5], 2021	Characteristics and Challenges of Low-Code Development: The Practitioners' Perspective	Studies what practitioners discuss about low-code and the challenges they report, such as customisation limits.	Supports the limitations in Section 8.4.
6	Di Ruscio et al. [6], 2022	Low-Code Development and Model-Driven Engineering: Two Sides of the Same Coin?	Relates low-code platforms to model-driven engineering.	Supports the data-model-first design in Section 4.2.
7	M. Hirzel [7], 2023	Low-Code Programming Models	Discusses the programming models used by low-code platforms.	Supports the no-code to pro-code spectrum in Section 8.5.
8	Zoho Corporation [8]	Deluge Script Reference Guide	Documents Deluge statements, criteria fetches, alerts and sendmail.	Reference for the seven scripts in Section 5.5.

2.5 Emerging Trends in Low-Code Development

Three trends bear on this application. The first is citizen development, in which people who are not professional programmers build and maintain departmental applications; a lost and found register is a typical case, because the person who understands the process can also change the application. The second is AI-assisted building, where an assistant proposes fields, scripts or whole applications from a description. Such output still has to be checked by the developer, and no Zia or AI field has been built into this application. The third is hyper-automation, in which routine steps such as validation, notification and status change happen without human action. This trend shaped the seven workflows of Chapter 5, which take over the checking and notifying that a staff member would otherwise do by hand.

2.6 Proposed System

The proposed system is a Zoho Creator application that records lost items, found items and claim requests in three linked forms. Each limitation in Section 2.2 is answered in turn. A single record removes the scattered reports. Mandatory fields and a fixed set of categories bring consistency. Every claim stores the claimant, the proof offered, the date and the claim status, and a subform keeps the actions taken on it. Workflows email the administrator when a found item is reported and when a claim is submitted. Reports and a dashboard supply lists and summaries, and permission profiles keep non-administrators from editing or deleting records.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	The lost and found in-charge who owns the application	Review found items and claims, approve claims, add remarks, manage users and permissions, view all reports and the dashboard
Student / User	Any student or staff member who loses, finds or claims an item	Report lost and found items, submit claim requests, view records through the reports allowed to the Student profile

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall record a lost item with contact name, email, item name, category, date lost and location lost as mandatory fields.	Student, Admin	High
FR-02	The system shall record a found item with finder name, item name, category, date found and location found as mandatory fields.	Student, Admin	High
FR-03	The system shall reject a lost date, found date or claim date later than the current date.	All	High
FR-04	The system shall reject a lost item report that has the same item name, date lost and location lost as an existing report.	All	High
FR-05	The system shall allow a claim to be raised against a found item chosen from a lookup and shall keep the actions taken on the claim in a subform.	Student, Admin	High
FR-06	The system shall email the administrator when a found item is submitted.	Admin	Medium
FR-07	The system shall email the administrator when a claim request is submitted.	Admin	Medium
FR-08	The system shall set the status of a found item to Claimed when its claim is approved.	Admin	High
FR-09	The system shall show a dashboard with item and claim counts, a claim-status summary and links to add new records.	Admin, Student	Medium
FR-10	The system shall give the Admin profile full access to records and the Student profile view-only access through reports.	Admin, Student	High
FR-11	The system shall be usable on the web and in the Zoho Creator mobile app.	All	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports with up to a few hundred records shall open within a few seconds on a normal broadband connection.
NFR-02	Security	Users shall be able to open, edit and delete only what their permission profile allows.
NFR-03	Usability	The application shall be usable in a web browser and in the Zoho Creator mobile app, with required fields clearly marked.
NFR-04	Availability	The application shall be hosted on the Zoho cloud so that no local server is needed.
NFR-05	Maintainability	Forms, reports and workflows shall carry meaningful names so that another person can modify them.
NFR-06	Auditability	The actions taken on a claim shall be recorded in the claim subform with a date.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (trial edition, Development environment)
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome, Microsoft Edge or Mozilla Firefox (latest version)
Email	Zoho Creator email action; a Gmail inbox was used to receive notifications
Mobile	Zoho Creator mobile app on an Android phone
Client hardware	Any computer that runs a current browser, with an internet connection

3.5 Use Case Model

Figure 3.1 shows the two actors and the use cases of the application. Two key use cases are described in Tables 3.5 and 3.6.

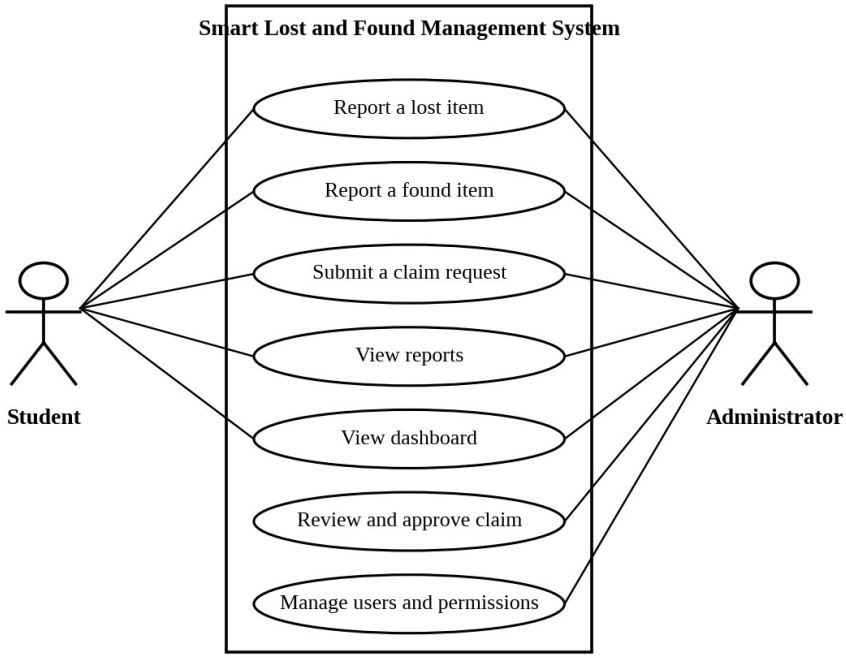


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description – Submit a Claim Request

Item	Description
Use Case ID	UC-01
Use Case Name	Submit a claim request for a found item
Actor(s)	Student
Pre-condition	The found item has already been recorded in the Found_Items form
Main Flow	1. The student opens the Claim Requests form. 2. The student enters name and email, selects the found item from the lookup, and enters the claim date and the proof of ownership. 3. The student submits the form. 4. The system validates the claim date, saves the record and emails the administrator.
Alternate Flow	If the claim date is later than today, the system shows “Claim Date cannot be in the future.” and cancels the submission.
Post-condition	The claim is saved with the status Pending
Related FR	FR-03, FR-05, FR-07

Table 3.6: Use Case Description – Approve a Claim

Item	Description
Use Case ID	UC-02
Use Case Name	Approve a claim and mark the item as claimed
Actor(s)	Administrator
Pre-condition	A claim request exists with the status Pending

Item	Description
Main Flow	1. The administrator opens the Claim Requests Report. 2. The administrator reviews the proof, edits the record, adds remarks and sets the Claim Status to Approved. 3. The system saves the record and sets the status of the linked found item to Claimed.
Alternate Flow	If the claim is not approved, the found item status is left unchanged.
Post-condition	The Found_Items Report shows the item as Claimed
Related FR	FR-08, FR-09

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

The solution has five layers, shown in Figure 4.1. Users reach the application through a web browser or the Zoho Creator mobile app. The application layer holds the three forms, the five reports, the dashboard page and the seven form workflows. An access-control layer of permission profiles decides what each user may open, edit or delete. Data is stored in the Zoho Creator cloud database, one table per form. Finally, the notification layer uses the built-in email action of Zoho Creator to send messages to the administrator's inbox. No external API is called.

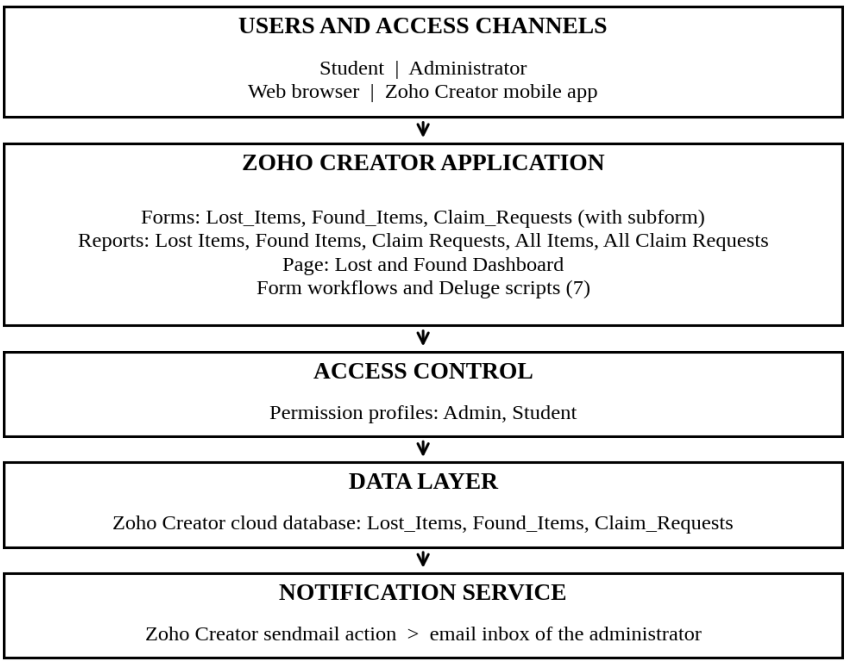


Figure 4.1: System Architecture of the Application

4.2 Data Model

The data model has three entities. Found_Items is the parent of Claim_Requests: each claim refers to exactly one found item through a lookup field, and one found item may receive several claims. Each claim owns a subform of action rows (action date, action taken and remarks), so a claim and its actions form a parent–child relationship. Lost_Items is a separate entity. Lost items are matched to found items by the administrator through the reports, because automatic matching is outside the scope of this version. Figure 4.2 shows the entities and Table 4.1 summarises the relationships.

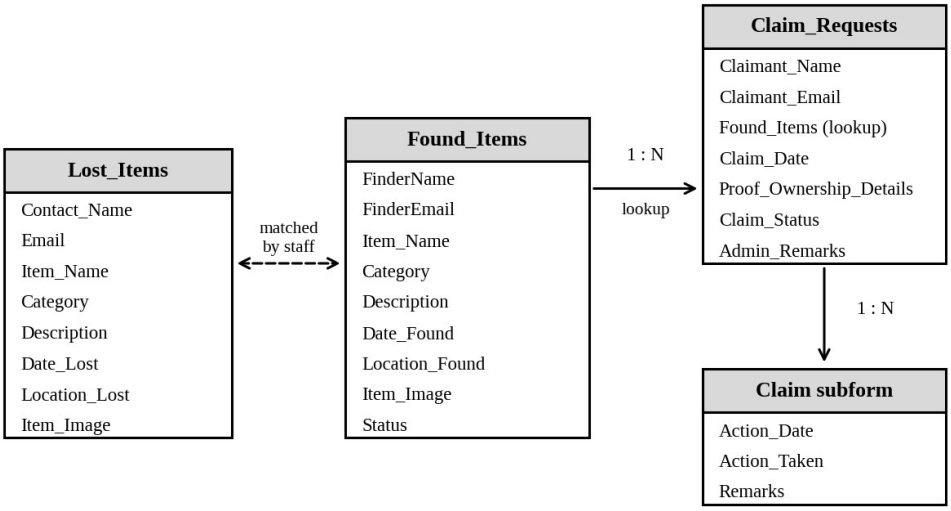


Figure 4.2: Entity–Relationship Diagram of the Application

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Found_Items	FinderName, Item_Name, Category, Date_Found, Status	Claim_Requests	One-to-Many	Lookup field Found_Items in Claim_Requests
Claim_Requests	Claimant_Name, Claim_Date, Claim_Status	Claim subform	Parent–Child	Subform
Lost_Items	Contact_Name, Item_Name, Date_Lost, Location_Lost	None (standalone)	Matched by staff through reports	No link field

4.3 Form Design

One field specification table is given for each form. Field types are named as they appear in Zoho Creator, and the script identifiers DS-01 to DS-07 refer to Table 5.2.

Table 4.2: Field Specification – Lost_Items Form

Field Label	Link Name	Field Type	Required	Validation / Default	Remarks
Contact Name	Contact_Name	Name	Yes	First and last name	Person who lost the item
Email	Email	Email	Yes	Valid email format	Contact for the owner
Item Name	Item_Name	Single Line	Yes	Used by DS-02	Part of duplicate check
Category	Category	Drop Down	Yes	Books, Electronics, Id Card, Wallet, Others	Fixed choices
Description	Description	Multi Line	No	—	Colour, brand, marks
Date_Lost	Date_Lost	Date	Yes	Not after today (DS-01)	Format dd-MMM-yyyy
Location_Lost	Location_Lost	Single Line	Yes	Used by DS-02	Part of duplicate check
Item_Image	Item_Image	Image	No	—	Photograph of the item

Table 4.3: Field Specification – Found_Items Form

Field Label	Link Name	Field Type	Required	Validation / Default	Remarks
FinderName	FinderName	Name	Yes	First and last name	Person who found the item
FinderEmail	FinderEmail	Email	No	Valid email format	Optional contact
Item_Name	Item_Name	Single Line	Yes	—	—
Category	Category	Drop Down	Yes	Books, Electronics, Id Card, Wallet, Others	Fixed choices
Description	Description	Multi Line	No	—	—
Date Found	Date_Found	Date	Yes	Not after today (DS-03)	Format dd-MMM-yyyy
Location Found	Location_Found	Single Line	Yes	—	—
Item_Image	Item_Image	Image	No	—	Photograph of the item
Status	Status	Radio	No	Set to Claimed by DS-07	Shown in Found_Items Report

Table 4.4: Field Specification – Claim_Requests Form

Field Label	Link Name	Field Type	Required	Validation / Default	Remarks
Claimant Name	Claimant_Name	Name	Yes	First and last name	Person making the claim
Claimant Email	Claimant_Email	Email	Yes	Valid email format	—
Found Items	Found_Items	Lookup	Yes	Lists records of Found_Items	Link to the parent form
Claim Date	Claim_Date	Date	Yes	Not after today (DS-05)	Format dd-MMM-yyyy
Proof/ Ownership Details	Proof_Ownership_Details	Multi Line	Yes	—	Description given as proof
Claim Status	Claim_Status	Drop Down	Yes	Default: Pending	Pending or Approved; read by DS-07
Admin Remarks	Admin_Remarks	Multi Line	No	—	Written by the administrator
SubForm	SubForm	Subform	—	Rows of Table 4.5	Action history of the claim

Table 4.5: Field Specification – Claim Subform

Field Label	Field Type	Required	Validation / Default	Remarks
Action Date	Date	Yes	Default: current date	Date of the action
Action Taken	Drop Down	Yes	Choice list, for example Claim Submitted	Step taken on the claim
Remarks	Multi Line	No	—	Free text note

4.4 Process and Workflow Design

The main automated process is the handling of a claim. When a claim is submitted, the claim date is validated first (DS-05). If the date is in the future the submission is cancelled. Otherwise the record is saved, an email is sent to the administrator (DS-06) and, if the claim status is

Approved, the linked found item is set to Claimed (DS-07). Figure 4.3 shows where each script runs. The same validate-then-notify pattern is used for the lost and found forms.

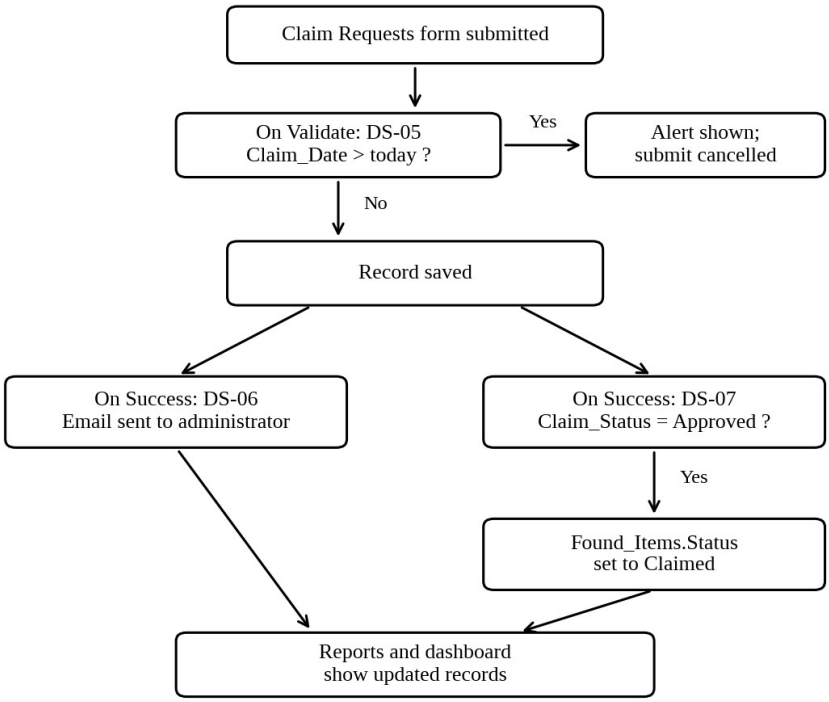


Figure 4.3: Workflow Process Flow for Handling a Claim Request

4.5 User Interface and Navigation Design

The application menu has four entries. The Items section holds the Lost Items form, the All Items report and the Lost Items Report. The Found_Items section holds the Found_Items form and its report. The Claim Requests section holds the form and two reports. The Lost and Found Dashboard page sits at the same level as the sections so that it is reachable with one click. Figure 4.4 shows this structure.

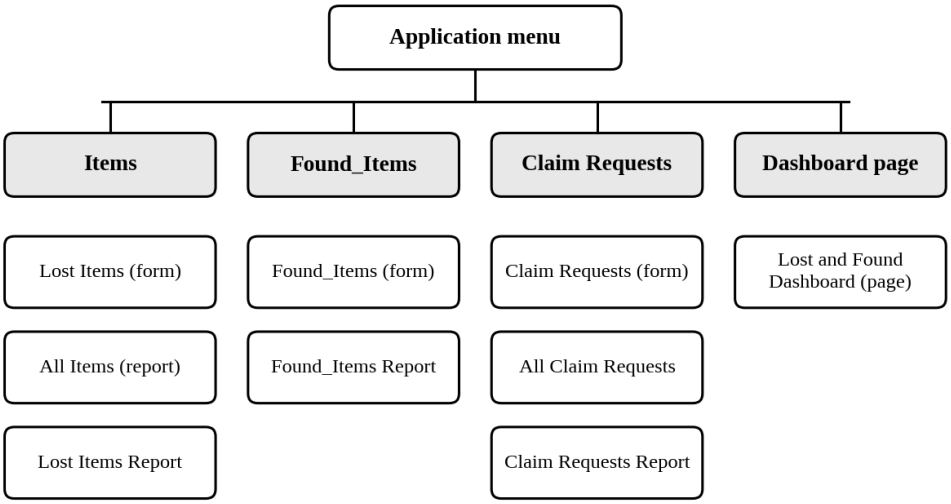


Figure 4.4: Navigation Structure of the Application

4.6 Access Control Design

Two permission profiles are defined. The Admin profile can open the three forms and can view, edit and delete records through the reports. The Student profile can open the three forms but can only view records through the reports; edit and delete are not granted. Table 4.6 states the matrix that Section 5.8 implements, using A = Add, E = Edit, V = View, D = Delete and — = no access.

Table 4.6: Role–Permission Matrix

Component	Admin Profile	Student Profile
Form: Lost_Items	A E V D	A V
Form: Found_Items	A E V D	A V
Form: Claim_Requests	A E V D	A V
Reports: Lost Items Report, All Items	V E D	V
Report: Found_Items Report	V E D	V
Reports: Claim Requests Report, All Claim Requests	V E D	V

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application Smart Lost and Found Management System was created in a Zoho Creator workspace and is worked on in the Development environment. Its menu has three sections (Items, Found_Items and Claim Requests) and one page (Lost and Found Dashboard). Figure 5.1 is the Workflow view of the application; it lists the three forms, the five reports, the seven workflows and the dashboard page in one place.

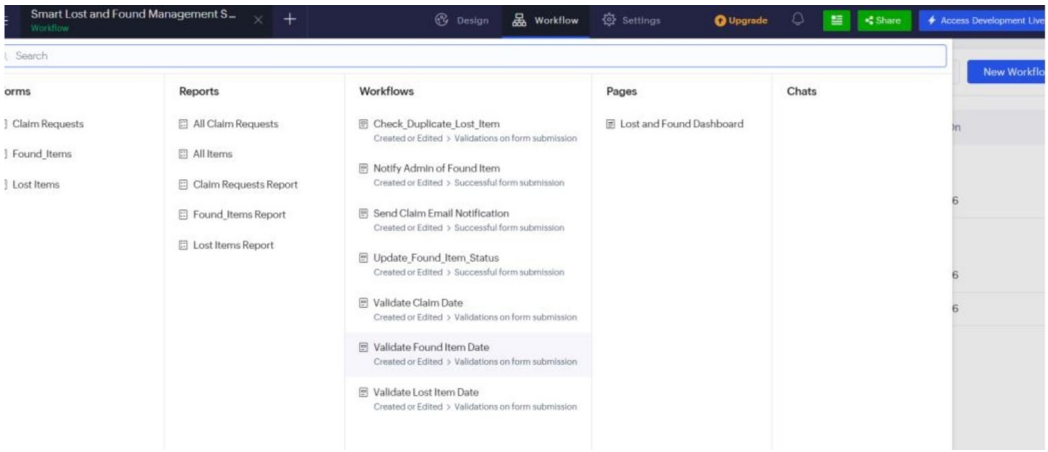


Figure 5.1: Application Components: Forms, Reports, Workflows and Pages

5.2 Forms and Data Capture

Ten field types are used across the forms: Name, Email, Single Line, Multi Line, Drop Down, Radio, Date, Image, Lookup and Subform. Mandatory fields are marked with a red asterisk, and Category and Claim Status are choice fields so that values stay consistent. The Lost_Items form (Figure 5.2) captures the owner's contact details and the lost item. The Found_Items form (Figure 5.3) captures the finder, the item and, later, its status.

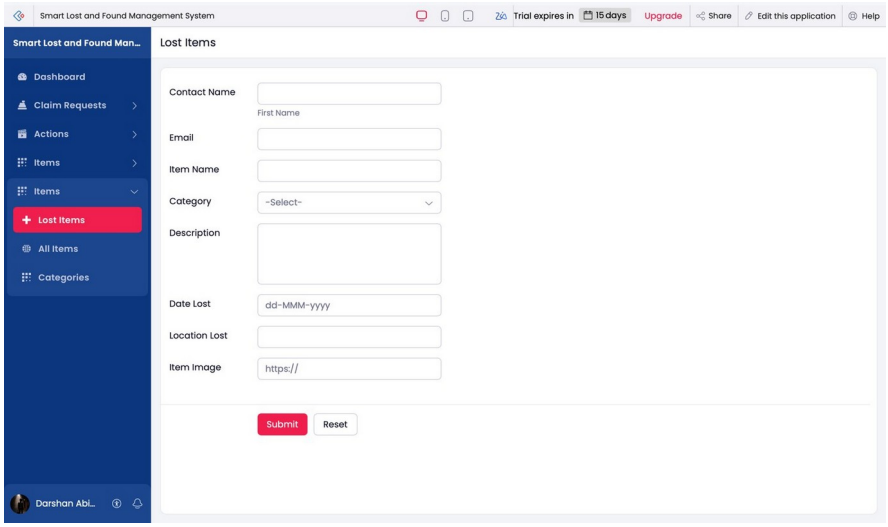


Figure 5.2: Lost_Items Form in Live Mode

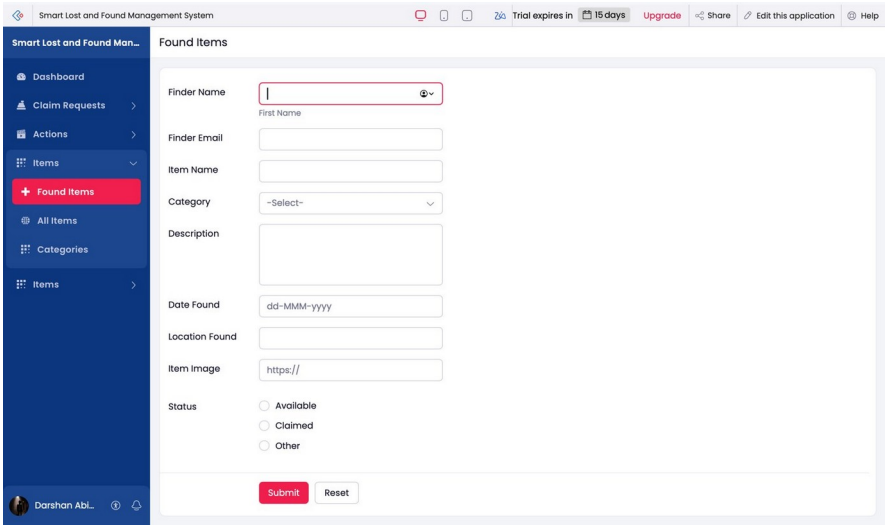


Figure 5.3: Found_Items Form in Live Mode

The Claim_Requests form (Figure 5.4) contains the lookup field Found Items, which lists the records of the Found_Items form, so a claim refers to an existing found item instead of repeating its details. Below the main fields is a subform whose rows record the action date, the action taken and remarks. Its first row is created with the current date, as Figure 5.4 shows.

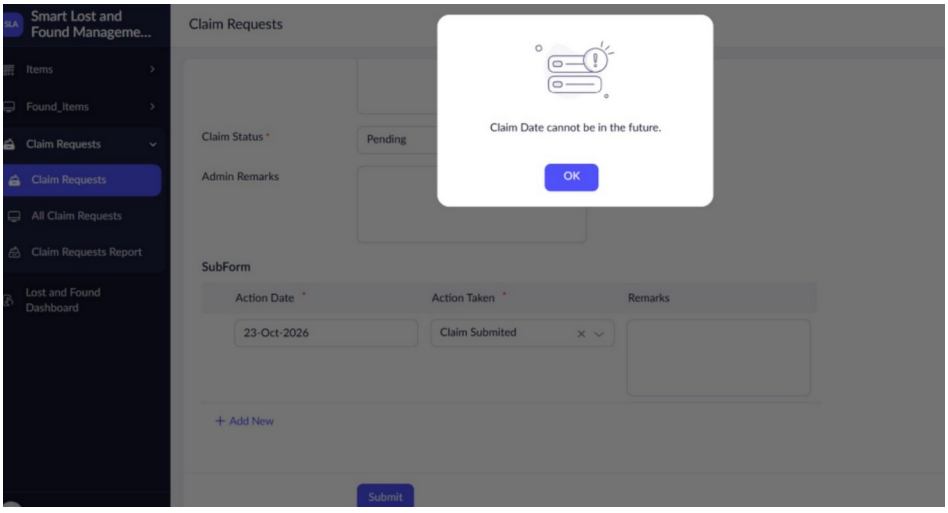


Figure 5.4: Claim_Requests Form with the Subform Visible and the Claim Date Validation Message

5.3 Reports

Five list reports were created. The main report of each form shows every field in a table. The two additional reports, All Items and All Claim Requests, give a second list view of the Lost_Items and Claim_Requests data. Table 5.1 lists them, and Figures 5.5 to 5.7 show the three main reports with the sample data entered for testing: 10 lost items, 11 found items and 12 claim requests.

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Section	Roles
Lost Items Report	List	Lost_Items	Items	Admin (view, edit, delete); Student (view)
All Items	List	Lost_Items	Items	Admin; Student (view)
Found_Items Report	List	Found_Items	Found_Items	Admin (view, edit, delete); Student (view)
Claim Requests Report	List	Claim_Requests	Claim Requests	Admin (view, edit, delete); Student (view)
All Claim Requests	List	Claim_Requests	Claim Requests	Admin; Student (view)

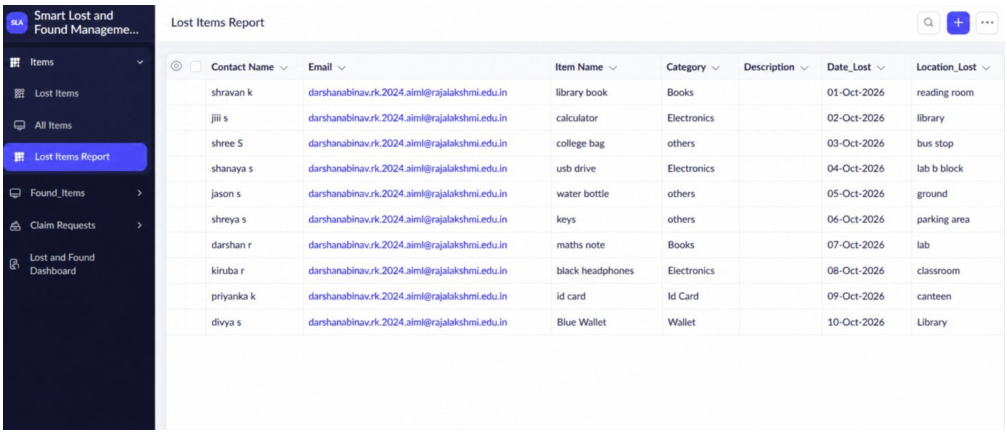


Figure 5.5: Lost Items Report

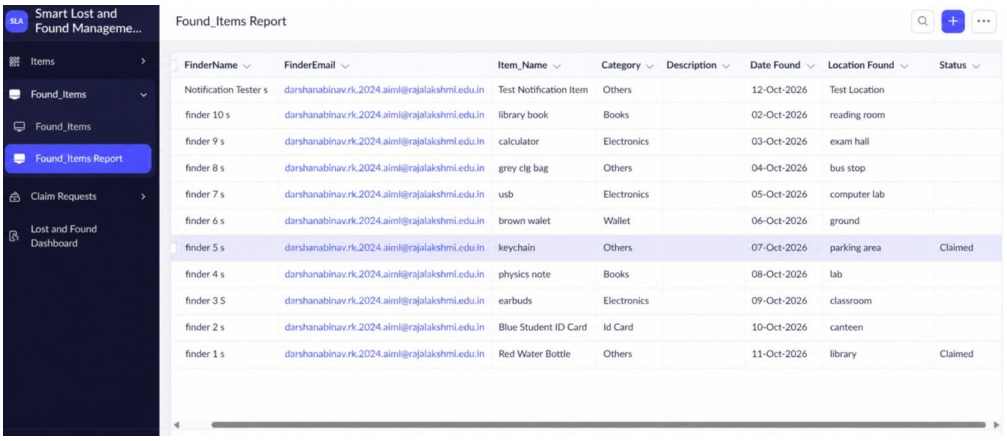


Figure 5.6: Found_Items Report Showing the Status Claimed

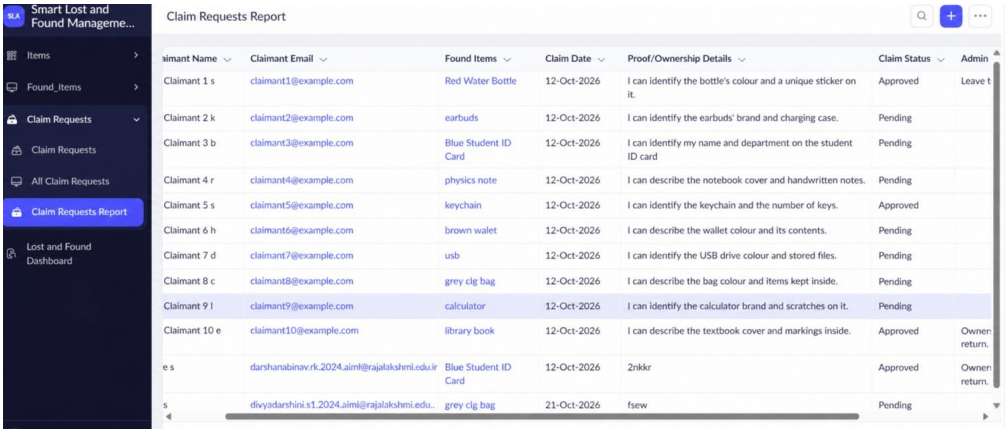


Figure 5.7: Claim Requests Report

5.4 Pages and Dashboards

The Lost and Found Dashboard is a page built in the Zoho Creator page builder. As captured in Figure 5.8, it shows a row of count tiles, a bar chart titled Distribution by Found Item, a Claim Status panel that counts approved, rejected, pending and total claims, and a Quick Links panel with shortcuts for adding a new claim request, claim action, found item or lost item. A second chart, Distribution by Claim Request, showed no data when the page was captured.

The figures on this dashboard (five claim requests, one approved, three rejected, one pending) do not agree with the twelve claim requests of Figure 5.7. This mismatch is recorded as BUG-03 in Table 6.2 and is treated as a defect that remains open; it is not hidden by choosing a different screenshot.

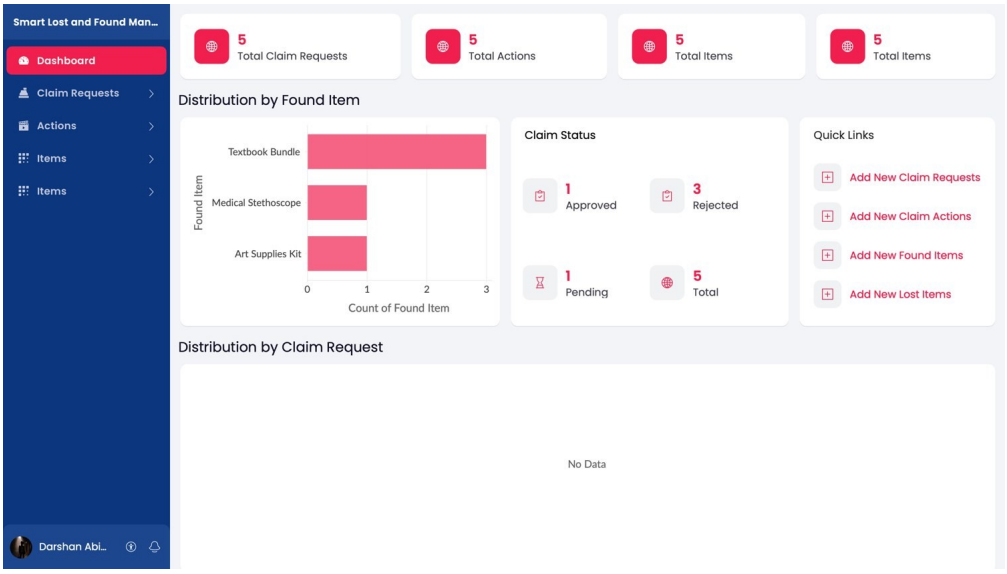


Figure 5.8: Lost and Found Dashboard

5.5 Deluge Scripting

Seven Deluge scripts are attached to form workflows. Four run when a form is validated and can stop the submission with cancel submit. Three run after a successful submission. Table 5.2 lists them with their location and the Deluge concepts used [8]. The complete code is in

Appendix C. No custom function, loop or collection is used, because the logic of this application does not need them.

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Lost_Items → On Validate (Validate Lost Item Date)	Reject a lost date in the future	if condition, zoho.currentdate, alert, cancel submit
DS-02	Lost_Items → On Validate (Check_Duplicate_Lost_Item)	Reject a duplicate lost report	criteria fetch, count(), alert, cancel submit
DS-03	Found_Items → On Validate (Validate Found Item Date)	Reject a found date in the future	if condition, zoho.currentdate, alert, cancel submit
DS-04	Found_Items → On Success (Notify Admin of Found Item)	Email the administrator about a found item	sendmail, zoho.adminuserid
DS-05	Claim_Requests → On Validate (Validate Claim Date)	Reject a claim date in the future	if condition, zoho.currentdate, alert, cancel submit
DS-06	Claim_Requests → On Success (Send Claim Email Notification)	Email the administrator about a claim	sendmail, zoho.adminuserid
DS-07	Claim_Requests → On Success (Update_Found_Item_Status)	Mark the found item as Claimed	if condition, criteria fetch by ID, field update

Code 5.1: DS-02 duplicate check on the Lost_Items form (On Validate)

```
// DS-02 | Form: Lost_Items | Event: On Validate
if(Lost_Items[Item_Name == input.Item_Name
   && Date_Lost == input.Date_Lost
   && Location_Lost == input.Location_Lost].count() > 0)
{
    alert "This lost item has already been reported.";
    cancel submit;
}
```

DS-02 counts the existing lost records that have the same item name, date and location. If the count is above zero a message is shown and the submission is cancelled. A single criteria count is used instead of fetching records and looping over them, which keeps the check light. Figure 6.1 shows the message.

Code 5.2: DS-05 claim date validation (On Validate)

```
// DS-05 | Form: Claim_Requests | Event: On Validate
if(input.Claim_Date > zoho.currentdate)
{
    alert "Claim Date cannot be in the future.";
    cancel submit;
}
```

DS-01 and DS-03 follow the same pattern for Date_Lost and Date_Found. In every case the user stays on the form with the values already entered, corrects the date and submits again.

Code 5.3: DS-06 claim notification email (On Success)

```
// DS-06 | Form: Claim_Requests | Event: On Success
sendmail
[
    from :zoho.adminuserid
    to :"darshanabinav.rk.2024.aiml@rajalakshmi.edu.in"
    subject :"New Claim Request Submitted"
    message :"A new claim request has been submitted. Please review
it in the Smart Lost and Found Management System."
]
```

Code 5.4: DS-07 status update of the found item (On Success)

```
// DS-07 | Form: Claim_Requests | Event: On Success
if(input.Claim_Status == "Approved")
{
    foundRec = Found_Items[ID == input.Found_Items];
    foundRec.Status = "Claimed";
}
```

DS-07 runs after a claim is saved. When the status is Approved it fetches the found record whose ID equals the value of the lookup field and sets its Status to Claimed. The result is visible in Figure 5.6 for the items Red Water Bottle and keychain; the behaviour for the other approved claims is discussed under BUG-02 in Section 6.4. DS-04 and DS-06 send a fixed message to one email address. The message produced by DS-06 is shown in Figure 6.2.

5.6 Workflows and Automation

All seven workflows are form workflows that run when a record is created or edited. They are shown in Figure 5.9 and described in Table 5.3. No schedule, approval, Blueprint, batch or report workflow was built, because the process does not need time-based or multi-stage routing at this scale.

Table 5.3: Workflows Implemented

Workflow Name	Type	Form	Trigger / Event	Action	Script
Validate Lost Item Date	Form workflow	Lost_Items	On validate	Block future date	DS-01
Check_Duplicate_Lost_Item	Form workflow	Lost_Items	On validate	Block duplicate report	DS-02
Validate Found Item Date	Form workflow	Found_Items	On validate	Block future date	DS-03
Notify Admin of Found Item	Form workflow	Found_Items	On successful submission	Send email	DS-04
Validate Claim Date	Form workflow	Claim_Requests	On validate	Block future date	DS-05
Send Claim Email Notification	Form workflow	Claim_Requests	On successful submission	Send email	DS-06

Workflow Name	Type	Form	Trigger / Event	Action	Script
Update_Found_Item_Status	Form workflow	Claim_Requests	On successful submission	Set status to Claimed	DS-07

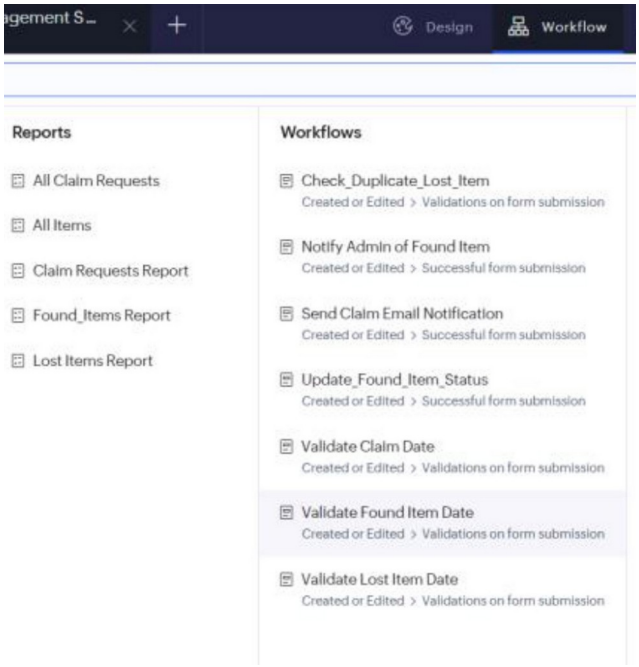


Figure 5.9: Workflows of the Application

5.7 Integrations

No connection, REST API call, webhook or Zoho CRM integration was used. The only outgoing communication is the built-in Creator email action, shown in Table 5.4. This keeps the application self-contained and avoids storing any credential. An external service would be added only for a requirement the platform cannot meet, for example sending the claimant a text message, which is listed as a future enhancement in Section 9.2.

Table 5.4: Integrations and Notifications

Item	Service / Endpoint	Method	Authentication	Purpose
Found item notification	Creator email	sendmail in DS-04	Platform account	Tell the administrator that an item was found
Claim notification	Creator email	sendmail in DS-06	Platform account	Tell the administrator that a claim was submitted
External API / CRM / webhook	Not used	—	—	—

5.8 User Management and Role-Based Access

The Users section of the application has three tabs: Permissions, Roles and Data Sharing. Two permission profiles were created (Figure 5.10). The Admin profile, described as allowing the administrator to work with all lost, found and claim forms and reports, has access to the three

forms and can view, edit and delete through the reports (Figure 5.11). The Student profile is described as allowing only verified access to view. It has access to the three forms and view access through the Claim Requests Report, Found_Items Report and Lost Items Report, while edit and delete are not ticked (Figure 5.12).

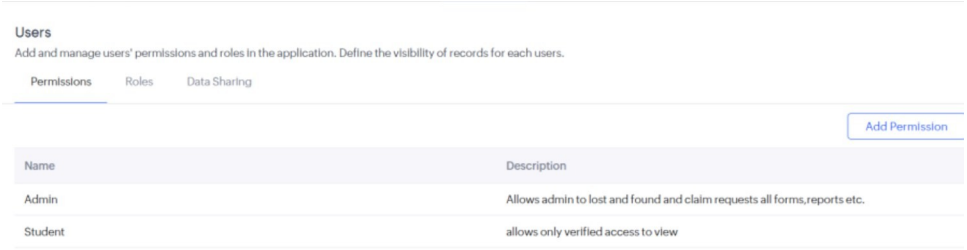


Figure 5.10: Permission Profiles Defined in the Application

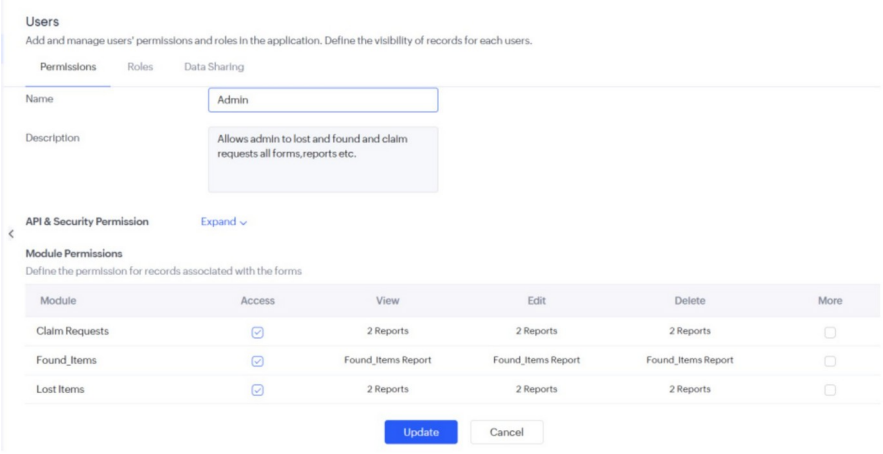


Figure 5.11: Admin Permission Profile

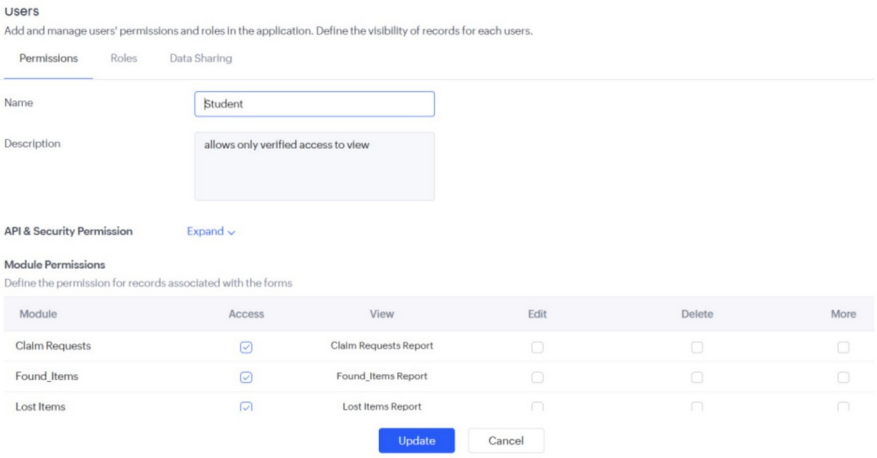


Figure 5.12: Student Permission Profile

Record-level criteria and a customer portal were not used. The behaviour of the Student profile was checked by reviewing its configuration; a login as a student user has not yet been carried out. This is stated again as a limitation in Section 8.4.

5.9 AI and Analytics Features

No Zia or AI field, prediction, OCR or sentiment feature is implemented in the application. The analytics feature is the dashboard of Section 5.4, whose tiles, bar chart and claim-status counts are computed from stored records. Their agreement with the reports was checked; the check found that the dashboard totals do not match the reports (BUG-03, Table 6.2), so the dashboard has to be corrected before it can be relied on.

5.10 Pro-Code Extension

No pro-code extension was used. Zoho Catalyst, JavaScript widgets or the Creator REST API were not needed, because every requirement was met by forms, reports, a page and short Deluge scripts. They would be considered if the system were extended to match lost and found items by text similarity or to send messages through an external gateway.

5.11 Mobile and Portal Access

The Zoho Creator mobile app opens the same application without extra development. Figure 5.13 shows the Lost Items form on an Android phone with a dark theme. The layout adapts to the narrow screen with one field per row, and the submit button stays at the bottom. A customer portal was not configured.

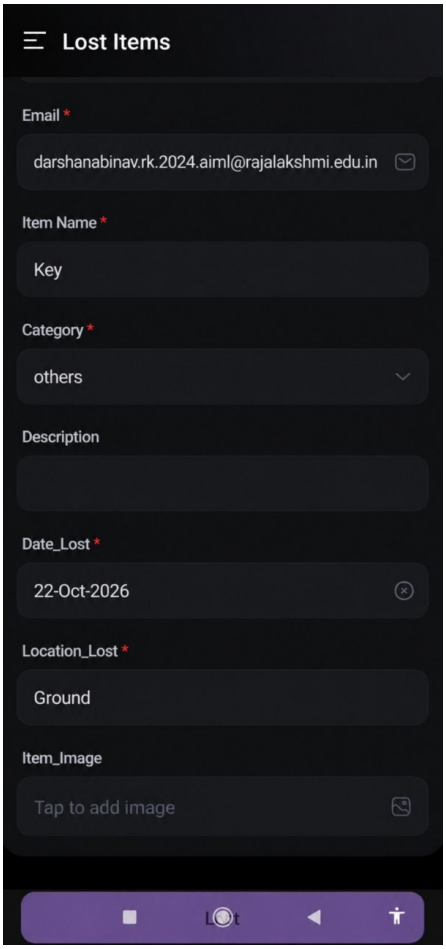


Figure 5.13: Lost Items Form in the Zoho Creator Mobile App

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was done in the Development environment at four levels. Unit testing checked each form with valid and invalid input. Script testing checked each Deluge script by triggering its event and observing the message or result. Integration testing followed a claim from submission to the status change of the found item, the email, the reports and the dashboard. Role-based testing was planned for each profile but was limited to a review of the profile configuration, as explained in Section 6.4. Sample data of 10 lost items, 11 found items and 12 claim requests was entered to exercise the reports and the dashboard.

6.2 Test Cases

Table 6.1 gives 17 test cases that cover every functional requirement in Table 3.2 and one non-functional requirement. A case marked Pass was executed and is supported by the figure named in the table. A case marked Partial or Fail was executed and did not fully meet its expected result. A case marked To verify has a defined procedure but its result has not yet been recorded.

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Submit a lost item with all mandatory fields	Fill all required fields and submit	Record saved	Records appear in Lost Items Report (Figure 5.5)	Pass
TC-02	FR-01	Submit a lost item without email	Leave Email empty and submit	Required-field error	—	To verify
TC-03	FR-02	Submit a found item with all mandatory fields	Fill required fields and submit	Record saved	Records appear in Found_Items Report (Figure 5.6)	Pass
TC-04	FR-03	Lost date in the future	Enter a Date_Lost after today	Alert shown; submit cancelled	—	To verify
TC-05	FR-03	Found date in the future	Enter a Date Found after today	Alert shown; submit cancelled	—	To verify
TC-06	FR-03	Claim date in the future	Enter a Claim Date after today	Alert shown; submit cancelled	Message “Claim Date cannot be in the future.” (Figure 5.4)	Pass
TC-07	FR-04	Duplicate lost report	Re-enter library book, 01-Oct-2026, reading room	Alert shown; submit cancelled	Message “This lost item has already been reported.” (Figure 6.1)	Pass
TC-08	FR-04	Same item at a different location	Use an existing item name and date with a new location	Record accepted	—	To verify
TC-09	FR-05	Claim against a found item with a subform row	Select a found item in the lookup and add an action row	Claim saved with the linked item	Claims show the linked found item (Figures 5.4 and 5.7)	Pass
TC-10	FR-06	Notification on a found item	Submit a found item	Email to the administrator	—	To verify

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-11	FR-07	Notification on a claim	Submit a claim	Email “New Claim Request Submitted”	Email received (Figure 6.2)	Pass
TC-12	FR-08	Approved claim updates the item	Set Claim Status to Approved	Found item status = Claimed	Claimed shown for Red Water Bottle and keychain, but not for library book and Blue Student ID Card, whose claims are also Approved (Figures 5.6, 5.7)	Partial
TC-13	FR-08	Pending claim leaves the item unchanged	Keep Claim Status Pending	Found item status unchanged	Items with pending claims have a blank status (Figure 5.6)	Pass
TC-14	FR-09	Dashboard counts	Compare dashboard figures with reports	Counts match records	Dashboard shows 5 claims; Claim Requests Report has 12 (Figures 5.7, 5.8)	Fail
TC-15	FR-10	Student profile permissions	Review the Student profile	View only; no edit or delete	Configuration matches (Figure 5.12)	Pass
TC-16	FR-11	Form on the mobile app	Open Lost Items in the mobile app	Form usable on a phone	Form displayed correctly (Figure 5.13)	Pass
TC-17	NFR-02	Student tries to edit a record	Log in as Student and edit	Access denied	Student login test not yet carried out	To verify

6.3 Debugging Deluge Scripts

Debugging relied on the messages produced by the scripts themselves and on checking results in the application. Each validation script shows an alert, so a wrong decision is visible on the form at once. For the email scripts a test record was submitted and the inbox was checked for the message; a test record named Test Notification Item remains in the Found_Items Report from such a check. For DS-07 the result was checked by opening the Found_Items Report after approving a claim, and that check is what exposed BUG-02. The Deluge editor was used to check syntax before a workflow was saved, and no script failed with a runtime error during testing.

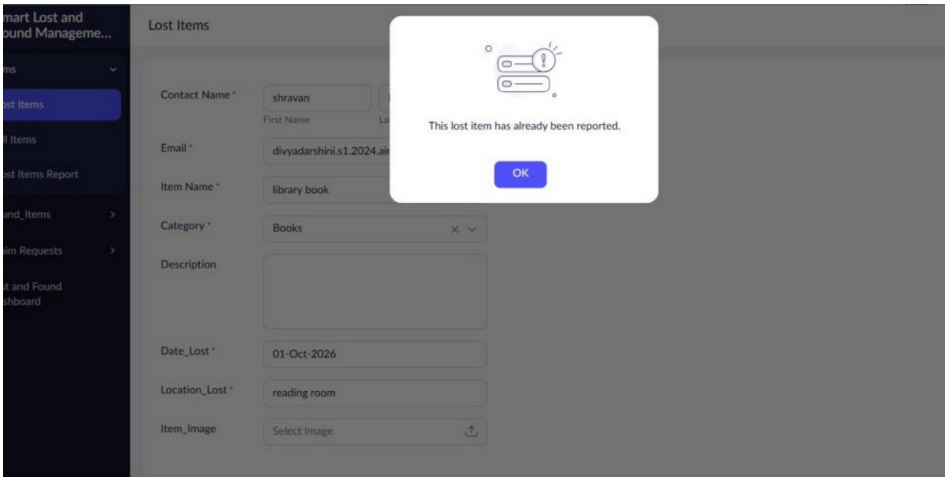


Figure 6.1: Duplicate Lost Item Message on the Lost_Items Form

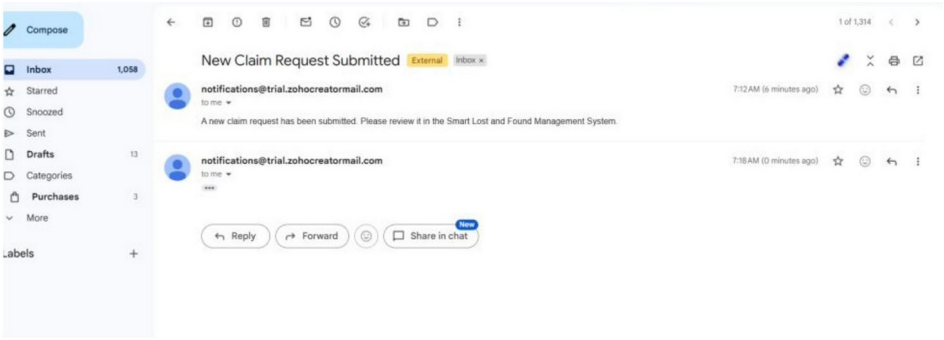


Figure 6.2: Claim Notification Email Received in the Inbox

6.4 Defect Log

Four defects were found. All are open at the time of writing, and the root cause is recorded only where it is known from the script itself.

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied / Planned	Status
BUG-01	The duplicate check can also stop an edit of an existing lost record that is saved without changes	DS-02 counts the record being edited as a match	Planned: add the condition ID != input.ID to the criteria	Open
BUG-02	Two found items, library book and Blue Student ID Card, are not marked Claimed although their claims are Approved	Not yet established	To investigate: re-save the approved claims and check whether DS-07 runs for edited records	Open
BUG-03	Dashboard totals (5 claim requests, 1 approved, 3 rejected, 1 pending) do not match the Claim Requests Report (12 claims)	Not yet established	To investigate the data source of each dashboard component and capture the page again	Open
BUG-04	The subform choice reads “Claim Submited”, a spelling error	Typing error in the choice list	Planned: correct the choice value	Open

The Student profile could not be exercised by a login test, so TC-17 remains To verify and the profile was checked by reviewing its configuration instead.

6.5 User Acceptance Testing

A user acceptance test with end users in each role has not yet been carried out. The application was walked through by its developer as an administrator, and the Student profile was reviewed as configured. Table 6.3 is therefore limited to that review.

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
Developer	Administrator	Forms, reports, validations and email work as designed; the dashboard totals and two item statuses need correction (BUG-02, BUG-03)	Defects logged in Table 6.2
Developer	Student (configuration review)	Profile allows view only through reports	Login test planned

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

Zoho Creator provides Development, Stage and Production environments. This project was built and tested in the Development environment only, as the label at the top of the live screens shows. The application has not been published to Production because the trial edition used for the project is close to expiry. For a real deployment the plan is to test in Stage with sample data, record release notes, publish to Production, re-create users and permission profiles there, and keep the previous version available for rollback. Table 7.1 states the present position as it is.

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases executed in Development / Stage	Partly (Table 6.1)
2	Sample/test data removed or isolated	No
3	Roles, permissions and users configured in Production	No (configured in Development only)
4	Connections authorised in Production	Not applicable (no connections)
5	Schedules and workflows enabled	Workflows enabled in Development; no schedules
6	Application published and URL verified	No
7	Version / release notes recorded	No

7.2 Performance Optimisation

The scripts avoid the two most common performance problems in Deluge. The duplicate check uses one criteria-based count and does not fetch records into a loop, and the status update fetches only the one found record whose ID equals the lookup value. No fetch sits inside a loop, because there are no loops. Reports show only the fields of their own form. The test data (about 33 records) is too small for a measurement to mean anything, so no before-and-after timing was recorded.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Single criteria count instead of fetch and loop	DS-02	Not measured	Not measured	Avoids reading every lost record
Fetch only the linked record by ID	DS-07	Not measured	Not measured	One record read per approval

7.3 Security Features

Security is provided mainly by the permission profiles of Section 5.8. In addition, the application keeps no password or API key in any script because no external service is called, and the emails are sent from the platform’s own notification address. Mandatory fields and the

validation scripts keep invalid data out of the database. Encrypted fields, IP restrictions and two-factor authentication were not configured.

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Permission profiles (Admin, Student)	Users section, all forms and reports	Unauthorised editing or deletion of records
Mandatory fields and Email field type	All forms	Missing or malformed contact data
Date and duplicate validation (DS-01, DS-02, DS-03, DS-05)	Form workflows	Invalid and repeated records
No stored credentials	All scripts	Credential leakage

7.4 Governance and Administration

Workflows and reports are named after what they do, such as Validate Claim Date and Update_Found_Item_Status, so that a successor can understand them. The claim subform keeps a dated trail of the actions taken on each claim, and the Admin Remarks field records the reason for a decision. The application has a single owner who is also the administrator. The Zoho Creator audit and backup features were not configured separately, and the test records still present in Development should be removed before any production use.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

A realistic scenario follows one item through the application. A finder hands in a red water bottle on 11-Oct-2026 and it is recorded in Found_Items (Figure 5.6). A claimant submits a claim against it, and the claim appears in the Claim Requests Report (Figure 5.7). The administrator reviews the proof and sets the claim to Approved, after which DS-07 sets the status of the found item to Claimed, as Figure 5.6 shows. The keychain found in the parking area followed the same path.

Across the sample data the Claim Requests Report holds 12 claims, of which eight are Pending and four are Approved. Two of the four approved claims are reflected as Claimed in the Found_Items Report; the other two are not (BUG-02). The dashboard of Figure 5.8 shows different totals from the reports (BUG-03). The results below should be read with these two defects in mind.

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Data model and forms	Sections 4.2, 4.3, 5.2	Three forms, a lookup and a subform (Figures 5.2 to 5.4) – achieved
Automation with Deluge and workflows	Sections 5.5, 5.6	Seven workflows; date validation, duplicate check and email observed (Figures 5.4, 6.1, 6.2); status update works for two of four approved claims – partly achieved
Reports and dashboard	Sections 5.3, 5.4	Five reports (Figures 5.5 to 5.7) achieved; dashboard built but its totals need correction (Figure 5.8) – partly achieved
Role-based access	Section 5.8	Admin and Student profiles configured (Figures 5.10 to 5.12); login test pending
Web and mobile use	Section 5.11	Mobile form view (Figure 5.13) – achieved
Testing and honest documentation	Chapters 6 and 7	Test cases, defect log and deployment checklist – achieved

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Recording an item or claim	Informal, often not written down	One form with mandatory fields	Every item and claim is recorded
Invalid or repeated data	Not checked	Date and duplicate checks	Bad data stopped at entry
Informing the administrator	By word of mouth	Email on each found item and claim	No manual informing

Parameter	Existing System	Proposed Application	Improvement
Updating item status	By hand, if at all	Automatic on approval (with the open defect BUG-02)	Fewer manual steps
Reports and counts	Not available	Five reports and a dashboard	Visibility, once BUG-03 is fixed
Data access	Anyone with the register	Permission profiles	Edit and delete restricted

8.4 Limitations of the Low-Code Approach

- **Trial and environment limits.** The application runs in the Development environment of a trial edition and has not been published, so the Student profile has not been demonstrated by login.
- **Fixed email recipient.** DS-04 and DS-06 send to one address written in the script. Sending to the claimant or to the address in the record would need a field reference and is a planned change.
- **Duplicate check on edit.** DS-02 can match the record being edited. Adding ID != input.ID to its criteria is the planned workaround (BUG-01).
- **Open defects.** Two approved claims are not reflected on their found items (BUG-02) and the dashboard totals disagree with the reports (BUG-03).
- **No automatic matching.** Lost and found items are matched by the administrator through the reports, because similarity matching is beyond what simple criteria can express.
- **Sender address.** In the trial, notification emails come from a Zoho trial notification address, as seen in Figure 6.2.
- **Vendor lock-in and customisation limits.** The application depends on Zoho Creator features, and a design beyond the components of the page builder would need a pro-code extension.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Most of the application is no-code configuration: the three forms and their fields, the lookup and subform, the five reports, the dashboard page and the permission profiles. The low-code part is the seven Deluge scripts, each of fewer than ten lines, which add validation, notification and status change. Nothing in this version needed pro-code. The point at which the project would move to pro-code is the matching of lost items with found items: comparing descriptions, categories and locations by similarity, or reading an identity card by OCR, would be better done in a custom function or in a Catalyst service called from Creator.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Smart Lost and Found Management System replaces scattered, unrecorded handling of lost items with one application on Zoho Creator. It has three forms with a lookup and a subform, five reports, a dashboard, seven Deluge workflows and two permission profiles, and it works on the web and in the mobile app. Date and duplicate checks keep the data clean, the administrator is notified by email, and an approved claim can update the status of the found item.

The project also shows the limits of the version submitted. It has been tested in the Development environment only, a student login test and a user acceptance test remain to be done, and matching of lost and found items is still manual. Four defects are open, among them two approved claims that are not reflected on their found items and a dashboard whose totals disagree with the reports. These are recorded in Chapters 6 to 8 so that the report matches what an examiner will see in the live application.

9.2 Future Enhancements

- Correct the open defects BUG-01 to BUG-04 and repeat the failed and partial test cases.
- Match lost items with found items automatically using category, date and location, and use Zia for description similarity.
- Send the notification email to the claimant and the finder, using the email fields of the record instead of a fixed address.
- Add an approval process or Blueprint so that a claim moves through submitted, verified, approved and handed-over stages.
- Add a scheduled workflow that reminds the administrator of found items that remain unclaimed for a set number of days.
- Add record-level criteria and a customer portal so that a student sees only their own reports and claims.
- Publish the application to Production after Stage testing and move the notification to a verified sender address.

9.3 Learning Outcomes and Career Relevance

Building this application gave practice in data modelling with lookups and subforms, form and report design, Deluge scripting with criteria fetches, workflow automation, permission design and testing. These skills match the work of a citizen developer or a low-code developer, and the habit of recording defects and test evidence as they are found is the one a solutions consultant or an automation engineer needs. The project also showed where low-code ends and pro-code begins, which is the judgement needed to move between those roles.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Each feature is marked Y only where it is implemented in the application and can be shown in the live demonstration. Features that were not built are marked N with the reason.

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created and sections/menu organised	Y	Sections Items, Found_Items, Claim Requests; Dashboard page	5.1	Figure 5.1
II	Forms with 8 or more distinct field types	Y	Lost_Items, Found_Items, Claim_Requests (10 types)	5.2	Figures 5.2–5.4
II	Lookup field (relationship between forms)	Y	Claim_Requests → Found_Items	4.2, 5.2	Figure 5.4
II	Subform (parent–child data)	Y	Claim_Requests subform	4.2, 5.2	Figure 5.4
II	Field validations / layout rules	Y	Date and duplicate validations; layout rules not used	5.2, 5.5	Figures 5.4, 6.1
II	List report with filter, grouping, custom action	N	Five list reports without grouping or custom actions	5.3	—
II	Kanban / Calendar / Timeline / Map report	N	Not built	5.3	—
II	Pivot table / chart report	N	Charts are on the dashboard page, not in a report	5.3, 5.4	—
II	Page / dashboard built with UI Builder	Y	Lost and Found Dashboard	5.4	Figure 5.8
II	Deluge – form events (validate / success)	Y	DS-01 to DS-07	5.5	Codes 5.1–5.4
II	Deluge – functions, fetch, loops, collections	N	Criteria fetch in DS-02 and DS-07; no custom function, loop or collection	5.5	Codes 5.1, 5.4
II	Email / in-app notifications	Y	DS-04, DS-06	5.5, 5.6	Figure 6.2
II	Form workflow	Y	Seven form workflows	5.6	Figure 5.9
II	Scheduled workflow	N	Not needed at this scale	5.6	—
II	Approval process	N	Claim status is set by the administrator	5.6	—
II	Blueprint	N	Not built	5.6	—
II	Batch / report workflow	N	Not built	5.6	—
II	Testing and debugging evidence	Y	Table 6.1; messages and email	6	Figures 5.4, 6.1, 6.2
III	Integration with a Zoho service	N	Email through the Creator action only	5.7	—
III	Connection / REST API call (invokeurl)	N	Not used	5.7	—
III	Webhook or Creator REST API	N	Not used	5.7	—
III	Users, roles and permission profiles	Y	Admin and Student profiles	5.8	Figures 5.10–5.12

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
III	Record-level access criteria / customer portal	N	Not used	5.8	—
III	Zia / AI feature	N	Not used	5.9	—
III	Analytics (charts / Zoho Analytics)	Y	Dashboard tiles, bar chart and claim-status counts (totals to be corrected, BUG-03)	5.9	Figure 5.8
IV	Performance optimisation applied and measured	N	Applied in DS-02 and DS-07; not measured	7.2	—
IV	Deployment across environments	N	Development environment only	7.1	—
IV	Security controls (encryption, OAuth, permissions)	Y	Permission profiles and validations	7.3	Figures 5.11, 5.12
IV	Governance (audit trail, administration)	N	Naming and subform trail; audit trail not configured	7.4	—
V	Pro-code extension (Catalyst / widgets / functions)	N	Not used; justified in 5.10	5.10	—
V	Low-code limitations and spectrum analysis	Y	Sections 8.4 and 8.5	8.4, 8.5	—
—	Mobile app / portal access	Y	Zoho Creator mobile app	5.11	Figure 5.13

Total features implemented: 15 / 32

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	2	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	3	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	2	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	1	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	2	

Levels: 1 – Low, 2 – Medium, 3 – High. The examiner may revise the self-assessed level.

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Lost_Items – Validate Lost Item Date

```
// Script ID : DS-01
// Location  : Lost_Items → On Validate
// Purpose   : Reject a lost date in the future
if(input.Date_Lost > zoho.currentdate)
{
    alert "Date Lost cannot be in the future.";
    cancel submit;
}
```

C.2 Lost_Items – Check_Duplicate_Lost_Item

```
// Script ID : DS-02
// Location  : Lost_Items → On Validate
// Purpose   : Reject a duplicate lost report
if(Lost_Items[Item_Name == input.Item_Name && Date_Lost ==
input.Date_Lost && Location_Lost == input.Location_Lost].count() >
0)
{
    alert "This lost item has already been reported.";
    cancel submit;
}
```

C.3 Found_Items – Validate Found Item Date

```
// Script ID : DS-03
// Location  : Found_Items → On Validate
// Purpose   : Reject a found date in the future
if(input.Date_Found > zoho.currentdate)
{
    alert "Date Found cannot be in the future.";
    cancel submit;
}
```

C.4 Found_Items – Notify Admin of Found Item

```
// Script ID : DS-04
// Location  : Found_Items → On Success
// Purpose   : Email the administrator about a found item
sendmail
[
    from :zoho.adminuserid
    to :"darshanabinav.rk.2024.aiml@rajalakshmi.edu.in"
    subject :"New Found Item Reported"
    message :"A new item has been reported as found. Please review
it in the Smart Lost and Found Management System."
]
```

C.5 Claim_Requests – Validate Claim Date

```
// Script ID : DS-05
// Location  : Claim_Requests → On Validate
// Purpose   : Reject a claim date in the future
if(input.Claim_Date > zoho.currentdate)
{
    alert "Claim Date cannot be in the future.";
    cancel submit;
}
```

C.6 Claim_Requests – Send Claim Email Notification

```
// Script ID : DS-06
// Location  : Claim_Requests → On Success
// Purpose   : Email the administrator about a claim
sendmail
[
    from :zoho.adminuserid
    to :"darshanabinav.rk.2024.aiml@rajalakshmi.edu.in"
    subject :"New Claim Request Submitted"
    message :"A new claim request has been submitted. Please review
it in the Smart Lost and Found Management System."
]
```

C.7 Claim_Requests – Update_Found_Item_Status

```
// Script ID : DS-07
// Location  : Claim_Requests → On Success
// Purpose   : Mark the found item as Claimed when the claim is
approved
if(input.Claim_Status == "Approved")
{
    foundRec = Found_Items[ID == input.Found_Items];
    foundRec.Status="Claimed";
}
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

- Open the Zoho Creator application link in Google Chrome, Microsoft Edge or Mozilla Firefox and sign in with the Zoho account that has been given access.
- On a phone, install the Zoho Creator mobile app, sign in with the same account and open Smart Lost and Found Management System.
- The menu on the left shows Items, Found_Items, Claim Requests and Lost and Found Dashboard.

D.2 Instructions for the Administrator

- Open Lost and Found Dashboard to see the item and claim summaries and the shortcuts for adding records.
- Open Found_Items Report to see found items and their status. An item shows Claimed after its claim is approved.
- Open Claim Requests Report, open a pending claim, read the proof, write Admin Remarks and change Claim Status to Approved to approve it.
- Check your email for the messages New Found Item Reported and New Claim Request Submitted.
- Open the Users section to review the permission profiles.

D.3 Instructions for the Student

- To report a lost item, open Items, then Lost Items, fill every field marked with a red asterisk and submit.
- To report a found item, open Found_Items, fill the form and submit it.
- To claim a found item, open Claim Requests, choose the item in Found Items, enter the claim date and describe how you can prove ownership, then submit.
- Dates cannot be later than today, and a lost item already reported with the same name, date and location is rejected.
- Use the report entries in the menu to view records. Students cannot edit or delete records.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition and Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports and Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting and Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC and AI	Zoho/external integrations, connections, roles and permissions, Zia/analytics	10	
6	Testing, Deployment, Security and Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		100	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	Below 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner: _____ Signature of External Examiner: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name: DARSHAN ABINAV RK Register No.: 241501039

Department / Section: AIML Date of Viva: _____

Application URL Verified: Yes / No Demo Duration: _____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	100	

Examiners' Remarks:

_____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Name: _____ Name: _____